

Ten-Layer Deterministic-Envelope/Quotient Frontier Review

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Abstract

RH-262–RH-271 close the stated deterministic-target envelope problem while leaving the moving-cloud bridge and uniform quotient problem open. The batch certifies a boundary supremum, derives an exact all-order parity dictionary, gives a factorwise order-29 tail below 2.6624745×10^{-5} , proves $|a_n| < 48q_*^n$, and proves the sharp law $a_n/q_*^n \rightarrow 1$. We add an exact finite-dimensional separation theorem: a complete root-of-unity shell can hide from any prescribed finite trace head while changing the next moment arbitrarily. Thus finite matching and a sharp target envelope cannot replace a cloud coefficient bridge. On the quotient side, a sufficient contour-stability theorem is available but none of its four continuum hypotheses is archived. The five-obligation vector remains $(0, 0, 0, 1, 1)$, the complete-certificate count is zero, and Gates A–E remain false/open.

1 Batch coordinate and target normalization

The repository is the sole source for this review. The route coordinate inherited from RH-270 is

`deterministic_target_envelope_sharp_legal_head_bridge_
uniform_quotient_open_complete_zero.`

Let the Hardy-scaled deterministic numerator satisfy

$$\log G_H(z) = - \sum_{n \geq 2} \frac{a_n}{n} z^n, \quad r_H = \frac{17}{20}. \quad (1)$$

RH-262 certifies on the rational circle $S = 7/5$ that

$$\sup_{|z|=7/5} |\log G_H(z)| < 107.906078 < 108, \quad (2)$$

without angular sampling [3]. This closes the previously open boundary-constant component; it does not select a noisy spectral cloud.

2 The exact deterministic package

Write $\lambda = 1.678573510428322 \dots$ and let T denote the reduced operator appearing in the factorization. RH-263 proves, for odd $n \geq 3$,

$$a_n = \frac{(r_H \lambda)^{-n}}{1 + \lambda^{-n}}, \quad (3)$$

and, for $k \geq 1$,

$$a_{2k} = r_H^{-2k} \left[2 \operatorname{Tr}(T^k) + \frac{2\lambda^{-2k}}{1 + \lambda^{-k}} - \frac{\lambda^{-2k}}{1 - \lambda^{-2k}} \right]. \quad (4)$$

These are all-order identities. Their comparison with all 27 RH-253 rows through order 28 has maximum floating residual $6.442918843840156 \times 10^{-14}$, but that comparison is only a regression test and is not used to prove the identities [4].

Set

$$q_* = (r_H \lambda)^{-1} = 0.7008752258547757 \dots, \quad \rho_* = q_*^{-1} = 1.4267874838640739 \dots \quad (5)$$

RH-267 proves the unified deterministic envelope

$$|a_n| < 48q_*^n \quad (n \geq 2), \quad (6)$$

using residue-class trace-ideal bounds, not a fit [8]. RH-268 then proves the sharp asymptotic law

$$\frac{a_n}{q_*^n} \longrightarrow 1. \quad (7)$$

Consequently ρ_* is the exact logarithmic radius, no smaller geometric base is possible, and the absolute logarithmic series diverges at the critical radius [9].

The direct factorwise estimate is much sharper than applying the single constant 48 at the current finite head. RH-264 certifies

$$\sum_{n \geq 29} \frac{|a_n|}{n} < 0.000026624745, \quad \exp\left(\sum_{n \geq 29} \frac{|a_n|}{n}\right) - 1 < 0.000026625100. \quad (8)$$

RH-265 certifies corresponding interfaces at first omitted orders 13, 21, 29, 37, 45, 53, 61; only $N = 29$ is paired with the archived order-28 head, while the higher rows remain conditional interfaces [5, 6].

Theorem 2.1 (Deterministic package boundary). *Equations (2)–(8) give a certified all-order target package with exact geometric base and radius. They imply neither a moving-cloud envelope nor an identity between cloud coefficients and a_n .*

Proof. The positive statements are precisely the cited factorization, interval, and trace-ideal results. Their hypotheses and conclusions mention only the deterministic coefficients a_n . No family $\tau_{n,s}$ and no equality or limit $\tau_{n,s} \rightarrow a_n$ occurs in those statements, so such a bridge is not a logical consequence. The next section gives an operator-level sharp counterexample to any attempted finite-head inference. $\square$

3 Finite-head matching alone cannot create a cloud bridge

Theorem 3.1 (Root-of-unity shell separation). *Fix $N \geq 1$ and put $m = N + 1$ and $\omega = e^{2\pi i/m}$. For $t > 0$, define the m -dimensional normal operator*

$$B_t = t^{-1/m} \text{diag}(1, \omega, \dots, \omega^{m-1}). \quad (9)$$

Then

$$\text{Tr}(B_t^n) = 0 \quad (1 \leq n \leq N), \quad \text{Tr}(B_t^{N+1}) = (N+1)t^{-1}. \quad (10)$$

Hence a finite-dimensional trace shell can preserve any prescribed trace head through order N under direct sum while making the next trace moment unbounded as $t \downarrow 0$.

Proof. For every $n \geq 1$,

$$\text{Tr}(B_t^n) = t^{-n/m} \sum_{j=0}^{m-1} \omega^{jn}.$$

The geometric sum vanishes unless m divides n , in which case it equals m . No integer $1 \leq n \leq N = m - 1$ is divisible by m , whereas $n = m$ is; this gives (10). If A is any finite-dimensional operator, then $\text{Tr}((A \oplus B_t)^n) = \text{Tr}(A^n) + \text{Tr}(B_t^n)$, so the first N moments are unchanged and the next one has the stated freedom. $\square$

Corollary 3.2 (Exact finite-head firewall). *No theorem using only finitely many matched trace moments and a bound on the separate deterministic sequence can, without an additional structural or continuity hypothesis, imply a uniform moving-cloud trace envelope or an all-order coefficient bridge.*

Remark 3.3. The shell is a logical separation witness. It is not asserted to occur in the archived noisy family, and it does not prove that the actual family is nonuniform. For the current order-28 atlas, Theorem 3.1 uses a 29-root shell: orders 1–28 vanish and order 29 is the first visible moment. This makes precise why a finite anchor cannot be promoted to an all-order cloud theorem, the central firewall of RH-241 [1].

4 The quotient theorem and its non-activation

The archived quotient evidence remains finite. RH-266 records 23/23 contractive twelfth powers, zero one-step contractions, and nine missing archived endpoints. It proves that these samples alone do not imply continuum uniformity, while explicitly not proving actual nonuniformity [7].

RH-269 gives the positive sufficient route. If the noisy operators converge in $\mathcal{S}_2$, a common positively oriented contour isolates the same finite-rank spectral cluster with a uniform resolvent bound, and the limiting orthogonal quotient has a contractive power, then fixed-space quotient compressions converge in $\mathcal{S}_2$. Uniform constants K_m , $\eta_m < 1$, and L_r follow, and for $\eta_m R^m < 1$ one obtains the RH-246 tail estimate

$$\sum_{n \geq m} \frac{|\mathrm{Tr}(C_s^n)| R^n}{n} \leq \frac{K_m R^m}{m(1 - \eta_m R^m)} \sum_{r=0}^{m-1} L_r R^r. \quad (11)$$

The archive verifies none of the four continuum hypotheses, so this exact criterion is not activated [2, 10]. Even activation would close only the specified quotient-tail obligation; it would not supply a legal head or the coefficient bridge.

5 The ten layers

paper	layer	strict result	first open input
RH-262	boundary budget	$M_{7/5} < 108$; rigorous order-29 Cauchy tail	legal head, bridge, quotient tail
RH-263	parity anchor	exact odd/even formulas at every order	cloud identification
RH-264	direct tail	factorwise order-29 tail below 2.6624745×10^{-5}	legal matching head
RH-265	tail ladder	seven certified omitted-order interfaces	heads beyond order 28
RH-266	uniformity logic	finite samples do not imply continuum control	modulus or interval family
RH-267	unified envelope	$ a_n < 48q_*^n$ for all $n \geq 2$	moving-cloud envelope
RH-268	sharp law	$a_n/q_*^n \rightarrow 1$ and exact radius ρ_*	no cloud-rate conclusion
RH-269	quotient criterion	four continuum inputs imply uniform RH-246 constants	all four inputs, currently 0/4
RH-270	certificate ledger	vector $(0, 0, 0, 1, 1)$; complete count zero	head, bridge, quotient tail
RH-271	frontier review	exact finite-head separation and scoped synthesis	operator-level reopening input

6 Ledger, protocol, and Gate boundary

The five independent obligations are

(legal head, coefficient bridge, uniform quotient tail,
analytic target tail, certified target boundary constant).

$$\boxed{(0, 0, 0, 1, 1)}, \quad (12)$$

so exactly two of five are satisfied and the complete-certificate count is zero [11]. This is a current-component ledger, not a global nonexistence result for all selectors or operator constructions.

The companion review reads the nine result files RH-262–RH-270, checks every archived Gate flag and theorem boundary, and records 187 structured witness records under an explicit bookkeeping rule. It also records the exact 29-shell witness for the order-28 frontier. No long operator calculation is rerun and no finite regression is extrapolated.

gate	required object	status after RH-271
A	canonical intrinsic dynamical determinant and identification	false/open
B	time-oriented scattering or unitary completion	false/open
C	self-adjoint generator and intrinsic $T \log T$ law	false/open
D	von Mangoldt-weighted prime-power traces	false/open
E	equality with the completed-zeta divisor	false/open

The next admissible reopening input is a legal operator-derived anchored head with an exact cloud bridge, or the full RH-269 continuum package: $\mathcal{S}_2$ convergence, a common finite-rank isolating contour, a uniform resolvent bound, and a contractive limiting quotient power. Finite endpoint extensions without a modulus do not meet this threshold.

No section constructs a Hilbert–Polya operator, identifies Riemann zeros, proves equality with the completed-zeta divisor, or implies the Riemann Hypothesis.

References

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